Bisection Method

Step-1:	Find points a and b such that $a<b$ and $f(a)\cdot f(b)<0$.
Step-2:	Take the interval $[a,b]$ and find next value $x_0=a+b/2$
Step-3:	If $f(x_0)=0$ then x_0 is an exact root, else if $f(a)\cdot f(x_0)<0$ then $b=x_0$, else if $f(x_0)\cdot f(b)<0$ then $a=x_0$.
Step-4:	Repeat steps 2 & 3 until $f(x_i)=0$ or $\lvert f(x_i)\rvert \le$ Accuracy

Example-1

1. Find a root of an equation $f(x)=x^3-x-1$ using Bisection method

Solution:

Here $x^3-x-1=0$

Let $f(x)=x^3-x-1$

Here

x	0	1	2
$f(x)$	-1	-1	5

1st iteration :

Here $f(1)=-1<0$ and $f(2)=5>0$

$\therefore$ Now, Root lies between 1 and 2

$x_0=1+2/2=1.5$

$f(x_0)=f(1.5)=0.875>0$

2*nd* iteration :

Here $f(1)=-1<0$ and $f(1.5)=0.875>0$

∴ Now, Root lies between 1 and 1.5

$x_1=1+1.5/2=1.25$

$f(x_1)=f(1.25)=-0.29688<0$

3*rd* iteration :

Here $f(1.25)=-0.29688<0$ and $f(1.5)=0.875>0$

∴ Now, Root lies between 1.25 and 1.5

$x_2=1.25+1.5/2=1.375$

$f(x_2)=f(1.375)=0.22461>0$

4*th* iteration :

Here $f(1.25)=-0.29688<0$ and $f(1.375)=0.22461>0$

∴ Now, Root lies between 1.25 and 1.375

$x_3=1.25+1.375/2=1.3125$

$f(x_3)=f(1.3125)=-0.05151<0$

5*th* iteration :

Here $f(1.3125)=-0.05151<0$ and $f(1.375)=0.22461>0$

∴ Now, Root lies between 1.3125 and 1.375

$x_4=1.3125+1.375/2=1.34375$

$f(x_4)=f(1.34375)=0.08261>0$

6th iteration :

Here $f(1.3125)=-0.05151<0$ and $f(1.34375)=0.08261>0$

∴ Now, Root lies between 1.3125 and 1.34375

$x_5=1.3125+1.34375/2=1.32812$

$f(x_5)=f(1.32812)=0.01458>0$

7th iteration :

Here $f(1.3125)=-0.05151<0$ and $f(1.32812)=0.01458>0$

∴ Now, Root lies between 1.3125 and 1.32812

$x_6=1.3125+1.32812/2=1.32031$

$f(x_6)=f(1.32031)=-0.01871<0$

8th iteration :

Here $f(1.32031)=-0.01871<0$ and $f(1.32812)=0.01458>0$

∴ Now, Root lies between 1.32031 and 1.32812

$x_7=1.32031+1.328122=1.32422$

$f(x_7)=f(1.32422)=-0.00213<0$

9th iteration :

Here $f(1.32422)=-0.00213<0$ and $f(1.32812)=0.01458>0$

∴ Now, Root lies between 1.32422 and 1.32812

x_8=1.32422+1.32812/2=1.32617

$f(x_8)$=f(1.32617)=0.00621>0

10th iteration :

Here f(1.32422)=-0.00213<0 and f(1.32617)=0.00621>0

∴ Now, Root lies between 1.32422 and 1.32617

x_9=1.32422+1.32617/2=1.3252

$f(x_9)$=f(1.3252)=0.00204>0

11th iteration :

Here f(1.32422)=-0.00213<0 and f(1.3252)=0.00204>0

∴ Now, Root lies between 1.32422 and 1.3252

x_{10}=1.32422+1.3252/2=1.32471

$f(x_{10})$=f(1.32471)=-0.00005<0

Approximate root of the equation x^3-x-1=0 using Bisection method is 1.32471

n	***a***	***f*(*a*)**	***b***	***f*(*b*)**	***c*=*a*+*b*2**	***f*(*c*)**	**Update**
1	1	-1	2	5	1.5	0.875	*b*=*c*
2	1	-1	1.5	0.875	1.25	-0.29688	*a*=*c*
3	1.25	-0.29688	1.5	0.875	1.375	0.22461	*b*=*c*
4	1.25	-0.29688	1.375	0.22461	1.3125	-0.05151	*a*=*c*
5	1.3125	-0.05151	1.375	0.22461	1.34375	0.08261	*b*=*c*
6	1.3125	-0.05151	1.34375	0.08261	1.32812	0.01458	*b*=*c*
7	1.3125	-0.05151	1.32812	0.01458	1.32031	-0.01871	*a*=*c*
8	1.32031	-0.01871	1.32812	0.01458	1.32422	-0.00213	*a*=*c*

9	1.32422	-0.00213	1.32812	0.01458	1.32617	0.00621	$b=c$
10	1.32422	-0.00213	1.32617	0.00621	1.3252	0.00204	$b=c$
11	1.32422	-0.00213	1.3252	0.00204	1.32471	-0.00005	$a=c$

Example

1.Example $f(x)=2x3-2x-5$

Here

x	0	1	2
$f(x)$	-5	-5	7

2. Example $x=\sqrt{12}$

Here

x	0	1	2	3	4
$f(x)$	-12	-11	-8	-3	4

3. Example $x=3\sqrt{48}$

Here

x	0	1	2	3	4
$f(x)$	-48	-47	-40	-21	16

4. Example-5 f(x)=x3+2x2+x-1

Here

x	0	1
f(x)	-1	3